FINAL HACKATHON PROJECT IDEA

APU ShuttleMate

Smart Bus Booking, Geo Check-in & QR Boarding

Challenge: Smart Campus & Student Experience
Goal: Help students secure a shuttle slot and make boarding more organised.

1. The problem

APU shuttle buses can get crowded, especially in the morning and evening. Students may arrive at the pickup point but find the bus full. They may not know how many slots are still available. If someone cancels or does not show up, the unused slot may not be offered to another student quickly enough.

2. Our solution

ShuttleMate is a website where students can view bus schedules, book a slot, check in using their phone location, and scan a QR code when boarding. When a booking is cancelled or a student misses the check-in deadline, the slot becomes available again automatically.

3. Eight main features

01	Bus Schedule & Routes	See routes, departure times and bus status. For continuous shuttle services, show estimated arrival or bus order instead of a fixed departure promise.
02	Bus Slot Booking	Book an available bus. If it is full, choose the next bus.
03	Live Bus Capacity	See available slots, active bookings, checked-in students and boarded passengers.
04	Geo Check-in	When near the pickup point, confirm arrival using the phone's location before the deadline.
05	Self-Service QR Boarding	Scan your QR code at the bus entrance. It works once, and only for the correct trip.
06	Automatic No-Show Recovery	If a student does not check in before the deadline, the booking expires and the slot opens again.
07	Seat Alert + FCFS Claim	Students can receive an alert for an earlier bus and claim a newly opened slot if eligible.
08	Admin & Bus Dashboard	Admins manage trips and capacity. Bus staff monitor boarding and handle problems.

No traditional waiting list. When a bus is full, students book the next available one. They can also turn on Seat Alerts to try for a slot on an earlier bus if one becomes free.

4. How the system works

1. Choose a bus: Check the route, departure information, and available slots.
2. Book a slot: Select an available bus and receive a QR boarding pass.
3. Geo Check-in: When near the pickup point, press Check In and allow access to your phone location.
4. Recover unused slots: Cancelled bookings and missed check-ins release slots automatically.
5. Seat Alert: Interested students receive a notification. Eligible students can press Claim; the first valid claim processed by the server wins.
6. Board the bus: Scan the QR code at the bus entrance. The system validates it and updates the boarded passenger count.

5. Live demo for the judges

Bus A (08:00) is full. Rani books Bus B (08:30) and turns on a Seat Alert for Bus A. One student booked on Bus A misses the check-in deadline, so their slot opens. Rani, who has completed Geo Check-in, receives an alert and successfully claims the earlier bus. Her Bus B booking is released automatically. When she scans her QR code on Bus A, the live capacity updates. Scanning the same QR again is rejected.

6. Why this project stands out

Automatic no-show recovery: Unused slots return without manual admin work. Self-service boarding: Students scan their own QR codes. Seat Alerts: Students can move to an earlier bus without repeatedly refreshing the page. Connected system: Bookings, check-ins, QR scans and capacity all update together.

7. Technology and prototype limits

Frontend	React + Tailwind CSS
Backend	Python + FastAPI
Database	SQLite
Location	Browser Geolocation API (with permission)
QR Code	Python qrcode + html5-qrcode
Notifications	In-app alerts; updates by polling or WebSocket

Important: GPS can be inaccurate or faked, so a QR scan at the bus entrance is still needed to confirm boarding. Check-in deadlines must match real bus operations. For buses without fixed departure times, staff can start and close each boarding session. A manual backup is needed if GPS, internet or the scanner fails.

Conclusion: APU ShuttleMate is more than a bus timetable. It helps students book slots, check in, board using QR codes and quickly reuse available slots. For the hackathon, we will demonstrate it with sample data, not official APU passenger records.